

PROFESSIONAL SYLLABUS

Software Development Life Cycle for Beginners

A beginner-friendly course that walks students through the complete software development journey — planning, design, development, testing, deployment, and maintenance — using real-world examples and zero coding required.

MODULES

8 + Capstone

LEVEL

**Absolute
Beginner**

FORMAT

**Theory +
Activities**

OUTCOME

SDLC-Ready



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Course Description

Software powers everything from mobile apps and websites to banking systems and healthcare platforms. Before any software is built, organizations follow a structured process known as the **Software Development Life Cycle (SDLC)**.

This course introduces beginners to the complete SDLC, helping them understand how software projects are planned, designed, developed, tested, deployed, and maintained. Using simple real-world examples such as a **To-Do List App**, students will learn how software teams transform ideas into working products **without requiring any programming knowledge**.

Overall Learning Objectives

- ✓ Explain the purpose and phases of the Software Development Life Cycle.
- ✓ Understand the roles and responsibilities of software project teams.
- ✓ Gather and document software requirements.
- ✓ Create basic process diagrams, user stories, and software designs.
- ✓ Understand testing, deployment, and maintenance concepts.
- ✓ Participate effectively in software projects even without coding skills.

Prerequisites: None. This course is designed specifically for complete beginners with no prior programming or technical experience.

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Course Modules

MODULE 01 Introduction to Software & SDLC

WHY THIS MODULE MATTERS Before understanding how software is built, students must understand what software is and why organizations use a structured development process.

KEY CONCEPTS

- What is Software?
- Types of Software
- Software Development Overview
- Introduction to SDLC
- Benefits of SDLC
- Common SDLC Models

LEARNING OBJECTIVES

- Understand the purpose of software.
- Identify different types of software applications.
- Explain why SDLC is important.

ACTIVITIES

- Discuss apps used daily.
- Create a software usage map.
- SDLC phase matching exercise.

ASSESSMENT Short quiz and software case study discussion.

MODULE 02 **Project Planning & Problem Identification**

WHY THIS MODULE MATTERS Every successful software project begins by understanding a problem and defining project goals.

KEY CONCEPTS

- Identifying Business Problems
- Project Goals and Objectives
- Stakeholders
- Scope Definition
- Feasibility Study
- Risk Awareness

LEARNING OBJECTIVES

- Define project scope.
- Identify stakeholders.
- Evaluate project feasibility.

ACTIVITIES

- Analyze a real-world software problem.
- Create a project scope document.

ASSESSMENT Mini project proposal.

MODULE 03 Requirements Gathering & Analysis

WHY THIS MODULE MATTERS Poor requirements are one of the leading causes of software project failure.

KEY CONCEPTS

- Functional Requirements
- Non-Functional Requirements
- User Stories
- Requirement Documentation
- Requirement Prioritization

LEARNING OBJECTIVES

- Gather user needs.
- Convert needs into requirements.
- Write simple user stories.

ACTIVITIES

- Interview simulation.
- User story workshop.

ASSESSMENT Create requirements for a To-Do List application.

MODULE 04**System Design Fundamentals**

WHY THIS MODULE MATTERS Design acts as the blueprint that guides development teams.

KEY CONCEPTS

- What is Software Design?
- User Interface Design
- User Experience Basics
- Process Flow Diagrams
- Wireframes and Mockups

LEARNING OBJECTIVES

- Understand design planning.
- Create simple wireframes.
- Draw process diagrams.

ACTIVITIES

- Sketch application screens.
- Design user journeys.

ASSESSMENT Create wireframes for a mobile application.

MODULE 05**Software Development Concepts**

WHY THIS MODULE MATTERS Students should understand what developers do, even if they do not write code themselves.

KEY CONCEPTS

- What Happens During Development
- Frontend vs Backend Concepts
- Databases Explained Simply
- Collaboration Between Teams
- Version Control Concepts

LEARNING OBJECTIVES

- Understand how software is built.
- Recognize different technical roles.
- Explain the development process.

ACTIVITIES

- Software team role-play.
- SDLC workflow mapping.

ASSESSMENT Development process diagram.

MODULE 06**Software Testing & Quality Assurance**

WHY THIS MODULE MATTERS Testing ensures software works correctly before users access it.

KEY CONCEPTS

- Importance of Testing
- Types of Testing
- Test Cases
- Bug Reporting
- Quality Assurance Basics

LEARNING OBJECTIVES

- Understand software quality.
- Create simple test cases.
- Identify common software defects.

ACTIVITIES

- Test a sample application.
- Write bug reports.

ASSESSMENT Prepare a testing checklist.

MODULE 07**Deployment & Software Release Management**

WHY THIS MODULE MATTERS Once software is developed and tested, it must be delivered safely to users.

KEY CONCEPTS

- Software Deployment
- Release Planning
- User Training
- Documentation
- Monitoring After Release

LEARNING OBJECTIVES

- Understand deployment processes.
- Explain release activities.
- Recognize the importance of documentation.

ACTIVITIES

- Deployment planning exercise.
- Release checklist creation.

ASSESSMENT Software launch plan.

MODULE 08 Maintenance, Support & Career Pathways

WHY THIS MODULE MATTERS Software continues evolving long after launch.

KEY CONCEPTS

- Corrective Maintenance
- Adaptive Maintenance
- Perfective Maintenance
- User Feedback Management
- SDLC Career Opportunities

LEARNING OBJECTIVES

- Understand software maintenance.
- Analyze user feedback.
- Explore software industry roles.

ACTIVITIES

- Maintenance scenario analysis.
- Career mapping exercise.

ASSESSMENT Maintenance improvement proposal.

Final Capstone Project

CAPSTONE PROJECT TITLE

Designing a To-Do List Application Using SDLC

Students apply every SDLC phase end-to-end through a guided real-world project.

STUDENT WORKFLOW

1. Identify the problem.
2. Define project goals.
3. Gather requirements.
4. Create user stories.
5. Design wireframes.
6. Prepare testing scenarios.
7. Create deployment plans.
8. Present maintenance strategies.

FINAL DELIVERABLES

- Requirement Document
- User Stories
- Wireframes
- Test Cases
- Deployment Checklist
- Final Presentation

Suggested Learning Resources

Curated supplementary materials to reinforce every concept covered in class.

Videos

- SDLC Introduction Videos
- Software Project Management Basics
- UI / UX Design Fundamentals

Reading Materials

- Beginner SDLC Guides
- Requirement Gathering Articles
- Software Testing Fundamentals

Instructor Materials

- Presentation Slides
- Case Studies
- Sample Requirement Documents
- Wireframe Templates

Practice Activities

- Group discussions & role-plays
- Mock interviews & user-story workshops
- Hands-on wireframing & bug-report writing

Assessment & Grading

Final evaluation is based on a balanced mix of assignments, activities, case studies, and a capstone project.

COMPONENT	WEIGHTAGE
Module Assignments	30%
Class Activities	20%
Case Studies	20%
Final SDLC Project	20%
Total	100%

COMPONENT	WEIGHTAGE
Presentation	10%
Total	100%

Course Outcome

After completing this course, students will understand the complete software development process — **from idea to maintenance**. They will be able to participate in software projects as **business analysts, project coordinators, QA trainees, product support staff**, or future developers with a strong foundation in SDLC concepts.

Next Steps After This Course

Students can continue their journey with any of the following learning tracks:

[Programming Fundamentals](#)[Web Development](#)[UI / UX Design](#)[Software Testing](#)[Business Analysis](#)[Agile & Scrum](#)[Project Management](#)[Product Management](#)

Program Promise: This SDLC Fundamentals course empowers absolute beginners — including **non-technical learners, students, and career switchers** — to confidently understand, contribute to, and grow within the software industry. No prior programming experience required.

OBSCYRATECH

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Understand how — without writing a line of code.

Join Obscyratech's SDLC Fundamentals program and gain the language, mindset, and structured thinking needed to step into the software industry as a business analyst, QA trainee, project coordinator, or future developer.

GET IN TOUCH

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LOCATION

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PROGRAM

**SDLC Fundamentals
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